

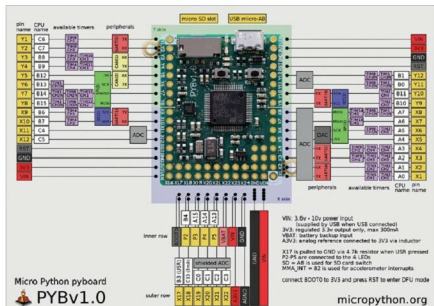


Fig. 2: The pyboard



Fig. 3: ESP8266 microcontroller chip

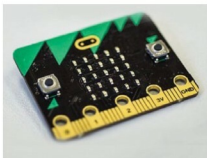


Fig. 4: BBC micro:bit

MicroPython support out-of-the-box. It features STM32F405RGT6 ARM Cortex M4 based processor. MicroPython supports APIs for almost all hardware peripherals of the chip like GPIO, I2C, SPI and UART.

ESP8266. MicroPython support for the low-cost Wi-Fi development ESP8266 microcontroller is excellent. Wi-Fi and Internet accessibility are made available within MicroPython using a special network library. There is also a Web based read-evaluate-print-loop (REPL) that allows you to run MicroPython code on ESP8266 using a Web browser within a local network.

BBC micro:bit. This is a new board aimed at teaching kids computer programming. Using MicroPython on micro:bit, you can access

the board's peripherals like LEDs and accelerometers.

Note. The full list of supported boards can be found at MicroPython website.

Library stacks supported by MicroPython

Python supports a variety of libraries on desktop computers, but porting these directly to a microcontroller is not an easy task as these are not optimised for running on machines with less RAM. Instead of making existing useful Python libraries available in MicroPython, these are stripped in order to be better optimised to run on constrained hardware. Current libraries supported by MicroPython are named with the initial u, which stands for Micro

Library implementation:

- ucollections - collection and container types
- uhashlib - hashing algorithm
- uheapq - heap queue algorithm
- uio - input/output streams
- ujson - JSON encoding and decoding
- uos - basic operating system services
- ure - regular expressions
- usocket - socket module
- ustruct - pack and unpack primitive data types
- utime - time-related functions
- uzlib - zlib decompression

Functionalities specific to MicroPython implementation are available in the following libraries:

- machine - for functions related to the board
- micropython - to access and control MicroPython internals
- network - for network configuration
- ctypes - to access binary data in a structured way

Following libraries are specific to pyboard:

- pyb - for functions related to the board
- Time-related functions
- Reset-related functions
- Interrupt-related functions

Note. The full list of supported libraries can be found at MicroPython website.

Loading MicroPython on ESP8266 board

ESP8266 is a cheap Wi-Fi development board. Since MicroPython exposes all required APIs including the networking stack, this chip is the preferred choice for learning how to load MicroPython. ESP8266 raw chip modules require a USB-to-TTL converter and a 3.3-volt stable power source to interact with the computer. Hence, you can use a complete ESP8266 board like NodeMCU or Adafruit Feather HUZZAH, which has all necessary components on the same board.